

IEEE Conference Presentation – Explanation & Q&A; Guide

Slide 1 – Title

Introduce yourself, your institution, and the paper title. Briefly mention the focus of the work and its relevance to sustainability and emission prediction.

Slide 2 – Motivation

Explain the importance of emission factor prediction for ESG and BRSR compliance. Highlight why traditional methods are insufficient and why a data-driven approach is needed.

Slide 3 – Problem Statement

Discuss the limitations of conventional emission estimation techniques such as static formulas and lack of real-time visualization or reporting integration.

Slide 4 – Proposed Solution

Describe the end-to-end machine learning framework using XGBoost and explain how the Streamlit dashboard supports ESG reporting.

Slide 5 – System Architecture

Walk through the system flow: data collection, preprocessing, feature engineering, model training, evaluation, and dashboard visualization.

Slide 6 – Why XGBoost

Justify the selection of XGBoost by explaining its ability to handle non-linear data, prevent overfitting, and provide higher accuracy.

Slide 7 – Feature Engineering

Explain the key features used such as energy consumption, production volume, fuel type, and operating hours, and how they improve model performance.

Slide 8 – Results and Performance

Explain the evaluation metrics used and compare the performance of XGBoost with Linear Regression and Random Forest models.

Slide 9 – ESG Dashboard

Explain how the dashboard provides real-time predictions, interactive visualizations, and automated ESG/BRSR reports.

Slide 10 – Conclusion and Future Work

Summarize the contributions and briefly discuss future enhancements such as IoT integration and additional sustainability indicators.

Expected Questions & Answers

Q: Why did you choose XGBoost?

A: XGBoost was chosen due to its strong performance with non-linear data, built-in regularization, and higher accuracy compared to traditional models.

Q: What data was used?

A: The model uses industrial and environmental data such as energy consumption, production volume, fuel type, operating hours, and historical emissions.

Q: How does this help ESG/BRSR compliance?

A: The framework generates emission predictions and converts them into ESG and BRSR-aligned reports using an interactive dashboard.

Q: How is your work different from existing methods?

A: Unlike traditional static approaches, this framework integrates machine learning prediction with real-time visualization and reporting.

Q: How do you handle missing data?

A: Missing values are handled using statistical imputation, and outliers are treated using the interquartile range method.

Q: What are the limitations?

A: The system depends on data quality and currently focuses mainly on emission factors, with scope to expand to other sustainability indicators.